

AWS-EnvBuilder

Human Integration and Operations Guide for Disposable Test and Development Environments

AUDIENCE	Application owners, developers, testers, technical project leads, and first-time AWS operators
BASELINE	Version 1.0; repository feature baseline 341f6fe, merged into main by 3aa0242
PURPOSE	Safe, repeatable, and reviewable AWS test/development environments

Prepared from the reviewed AWS-EnvBuilder repository. No AWS resource is created by reading this document.
Repository: <https://github.com/cbieger/AWS-EnvBuilder>

Contents and reading path

Read the sections in order for a first deployment. Returning operators can use the quick-reference and failure sections directly.

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CAUTION: A command is not permission

Examples show syntax. They do not replace account-owner approval, a reviewed Terraform plan, the exact runtime confirmation, or your organization's change process.

1. What this kit does - and does not do

NOTE: Plain-language summary

AWS-EnvBuilder creates a small, replaceable AWS web environment from reviewed Terraform. A load balancer sends HTTP requests to a Docker container on one EC2 instance; Auto Scaling may replace it or temporarily add a second instance. The instance is disposable, so the application must keep important data somewhere else.



Default AWS resources

- A dedicated VPC, two public subnets, an Internet Gateway, and security groups in one Region.
- An internet-facing Application Load Balancer on HTTP port 80 and an Auto Scaling Group with one t3.micro instance by default, scaling to at most two.
- An ECR repository for the application image and an immutable image digest for deployment.
- CloudWatch routine and error logs, plus an S3 bucket for short-lived load-balancer request logs.
- Three account-wide AWS Budgets at approximately US\$0.01, US\$1, and US\$5 for actual and forecast gross spend notifications.
- A protected, versioned S3 Terraform backend created separately so routine destruction cannot casually erase recovery history.
- An optional AWS-hosted scheduled self-destruct controller, notification path, and deletion-only build.

Deliberately outside the starter kit

- HTTPS/TLS, a domain name, WAF, private subnets with NAT, VPN access, databases, object-storage application data, backups, and production compliance controls.
- A promise that AWS usage is free, instantaneous billing alerts, or guaranteed delivery of SMS/email notifications.
- A safe place for passwords, API keys, customer data, uploads, server-side sessions, or irreplaceable files on the EC2 disk.
- Automatic Git commits, pushes, pull requests, or merges. Source control changes remain a separate reviewed human action.

STOP: HTTP is unencrypted

The default endpoint is a development stub on HTTP. Do not send credentials, cookies, personal data, or private business data through it. Add a reviewed domain, ACM certificate, HTTPS listener, and redirect before meaningful sensitive use.

2. Choose the integration layout

The supported distribution path is a source-only, checksummed archive. The destination application remains the parent project; the infrastructure kit sits below it in a clearly named folder.

Recommended destination layout

```
customer-application/  
  Dockerfile  
  .dockerignore  
  package.json          # example application manifest  
  src/  
  infrastructure/  
    aws-envbuilder/  
      README.md  
      docs/  
      scripts/  
      terraform/  
      tests/
```

Create a reviewed package at the source

1. Open a terminal at the top of the AWS-EnvBuilder source repository.
2. Run `./scripts/workspace.sh validate`. Stop on any nonzero exit.
3. Run `./scripts/package.sh --version 1.0.0` or use your approved release label.
4. Keep the generated `.tar.gz` and matching `.sha256` together.

```
./scripts/workspace.sh validate  
./scripts/package.sh --version 1.0.0  
  
# macOS verification  
cd dist  
shasum -a 256 -c aws-envbuilder-1.0.0.tar.gz.sha256  
  
# Linux verification  
sha256sum -c aws-envbuilder-1.0.0.tar.gz.sha256
```

Attach it to the destination project

```
mkdir -p /absolute/path/to/customer-application/infrastructure  
tar -xzf aws-envbuilder-1.0.0.tar.gz -C /absolute/path/to/customer-application/infrastructure  
mv /absolute/path/to/customer-application/infrastructure/aws-envbuilder-1.0.0 /absolute/path/to \  
/customer-application/infrastructure/aws-envbuilder  
cd /absolute/path/to/customer-application/infrastructure/aws-envbuilder  
./scripts/workspace.sh validate
```

READY CHECK: Inspect before committing

The package intentionally excludes Git history, AWS credentials, `.workspace`, `.terraform`, `real tfvars`, `backend.hcl`, Terraform state/plans, runtime logs, inventory reports, contact data, application files, and old release archives. Review the extracted file list and your Git diff before your team commits it.

Updating an attached copy

1. Create and verify a new package; do not unpack it over the live copy.

2. Extract beside the old folder and compare with your Git diff or `diff -ru`.
3. Preserve local ignored settings: `terraform.tfvars`, `backend.hcl`, `application.auto.tfvars.json`, `scheduled_destroy.auto.tfvars.json`, `.workspace`, `state`, `plans`, and `credentials`.
4. Replace only reviewed source files, validate locally, and run a fresh Terraform plan. Treat backend or state migration as a separate change with a backup.

3. Prepare the application before AWS

The application contract

Requirement	What good looks like	Why it matters
Docker build	Dockerfile with a pinned/versioned base image; all manifests and lock files present.	The same reviewed source should produce a predictable image.
Build exclusions	<code>.dockerignore</code> excludes at least <code>.git</code> and <code>.env*</code> , plus keys, caches, and local build output.	Docker must not receive secrets or unrelated workstation files.
Network	Process listens on 0.0.0.0 and the configured <code>container_port</code> .	The load balancer cannot reach a loopback-only listener.
Health	Fast unauthenticated route returns HTTP 200-399.	Auto Scaling uses it to decide whether an instance is safe for traffic.
State	No durable data, uploads, sessions, or irreplaceable files on local disk.	Instances may be replaced without warning.
Logs	Useful <code>stdout/stderr</code> ; errors to <code>stderr</code> ; no secrets or personal data.	CloudWatch collects streams and retains errors longer.
Shutdown	Tolerates interrupted connections and brief version overlap.	Rolling replacement can run old and new containers together.

Run the read-only application inspection

```
./scripts/inspect_app.sh /absolute/path/to/customer-application

# Machine-readable result for CI or an AI operator
./scripts/inspect_app.sh /absolute/path/to/customer-application --json
```

The scanner inventories common manifests, checks Docker safety files, warns about floating base tags/root containers/missing lock files, and blocks obvious private-key headers or AWS credential-shaped values. It never prints the suspected secret. A passing static scan is not a trust decision: review Dockerfile commands, source, install scripts, and base-image provenance.

Minimum owner decisions before proceeding

- Which port and health path the container actually uses.
- Where uploads, databases, sessions, and secrets will live if the application needs them.
- Which office/VPN IP ranges may reach the HTTP load balancer. Avoid 0.0.0.0/0 unless the owner accepts public reachability.
- Which AWS account, Region, project name, environment name, alert email, and cost owner are approved.

4. One-time workstation and AWS identity setup

STOP: Root means AWS account root

The first-run script requires an STS identity whose ARN ends in :root. It does not require the macOS/Linux root account and should not be run with sudo. After setup, ordinary AWS-facing commands fail when they detect AWS root unless an owner explicitly adds the one-command --run-as-root exception.

Workstation prerequisites

Program	Purpose	Operator check
AWS CLI v2	Authentication and read/write AWS API calls.	aws --version; update to the current patch.
Terraform	Creates, updates, and deletes declared resources.	terraform version
Python 3	Inspection, cost, inventory, schedule, and tests.	python3 --version
jq + curl	JSON checks and official-version/network checks.	jq --version; curl --version
Docker	Required only when building/pushing the application.	docker version

The strict preflight checks availability and AWS CLI currency. It does not silently install or upgrade programs. Follow docs/WORKSTATION_SETUP.md, reopen the terminal after installation, then rerun validation.

One-time identity handoff

```
./scripts/workspace.sh validate
aws login --profile aws-root-bootstrap
aws sts get-caller-identity --profile aws-root-bootstrap

./scripts/first_run_setup.sh --root-profile aws-root-bootstrap --region us-west-2 --service \
-account aws-envbuilder-automation

aws logout --profile aws-root-bootstrap
./scripts/workspace.sh preflight --region us-west-2
```

1. Before the setup command, write down the intended 12-digit AWS account ID. Stop unless STS returns exactly that account and an ARN ending in :root.
2. The helper shows the IAM user/local profile proposal and requires the phrase CREATE AWS SERVICE ACCOUNT.
3. It creates one tagged IAM user, one tagged customer-managed policy, and one long-term key. The key is written directly to the protected AWS shared-credentials file and is not displayed or logged.
4. The helper verifies the new user, performs strict read-only preflight, and stores only the non-secret profile name in ignored .workspace/service-account-profile.
5. End the root session. The final preflight must print an ARN ending in :user/YOUR_SERVICE_ACCOUNT, never :root.

CAUTION: Long-term credential tradeoff

The requested standalone service account uses a powerful long-term access key. AWS recommends short-lived roles or IAM Identity Center where practical. An organization should replace this bootstrap with an approved deployment role, permission boundary, account guardrails, and narrowly scoped tag/ARN conditions.

5. Configure the environment and protected state

Create the ignored settings file

```
cp terraform/terraform.tfvars.example terraform/terraform.tfvars
```

Setting	Required decision
project_name	3-24 lowercase letters/digits/hyphens; starts with a letter and ends with a letter or digit.
environment	3-12 lowercase letters/digits/hyphens, such as dev, test, qa, or demo.
aws_region	The approved enabled Region, such as us-west-2.
allowed_ipv4_cidrs	One or more trusted IPv4 CIDRs; narrow this to office/VPN ranges.
container_port + health_check_path	Exact listener port and unauthenticated 200-399 health route.
instance counts	minimum <= desired <= maximum; each is 1-4. Default is one, maximum two.
budget_alert_emails	One to ten monitored addresses. Never put email passwords/tokens here.
retention	Routine log value supported by CloudWatch; errors at least 30 days and longer than routine logs.

Create the protected Terraform backend

```
./scripts/bootstrap_backend.sh --region us-west-2 --project customer-demo --environment dev
```

This one-time write creates a deterministically named, encrypted, versioned S3 bucket and an ignored terraform/backend.hcl. It asks for exact confirmation. Normal destroy keeps the bucket so state history remains available for recovery and audit.

Prove readiness before any Terraform write

```
./scripts/workspace.sh preflight --region us-west-2  
./scripts/workspace.sh init --region us-west-2
```

- Preflight authenticates with STS, proves Region reads, and uses the IAM policy simulator for the required create/delete action set.
- An organization policy, permission boundary, service control policy, quota, or Region restriction may still block a later action. Policy simulation is evidence, not an absolute guarantee.
- Do not delete a Terraform lock object just because init reports a lock. First confirm no other operator or automation is using the state.

6. Understand costs before creation

STOP: THIS BUILD IS NOT GUARANTEED TO BE FREE

The repository's conservative low-traffic estimate for the default us-west-2 workspace is about US\$1.40 per day, or about US\$42 per 30-day month, before tax, heavy traffic, unusual logs, data transfer, scaling, SMS registration, or price changes. Verify AWS prices and your account's Free Tier status immediately before apply.

What usually drives the bill

Cost area	Why it appears	Control
Application Load Balancer	Hourly base plus load-capacity usage.	Destroy idle environments; reduce traffic; do not add unnecessary ALBs.
EC2 + EBS	Instance hours and encrypted root storage; Auto Scaling can run two.	Keep desired/minimum low; review maximum; consider Spot only if interruptions are acceptable.
Public IPv4	The instance uses a public address for outbound access while inbound is blocked.	Destroy when idle; redesign with endpoints/private networking only after comparing costs.
Logs + S3	Ingestion, retained logs, ALB objects, and state/source versions.	Keep routine retention short; retain errors longer; never enable noisy debug logs indefinitely.
Budgets	Three account-wide budget definitions and notifications.	Treat notifications as delayed alarms, not automatic spending stops.
Scheduled teardown	Dedicated toll-free number, SMS, Scheduler, Lambda, DynamoDB, SNS, CodeBuild, logs, S3.	Use only when the deadline control is worth registration and recurring fees.

Cost controls that must remain enabled

- Enter monitored budget email addresses and confirm every AWS Budgets subscription message.
- Enable AWS native Receive AWS Free Tier alerts under Billing preferences. The near-zero budget is not a service-by-service Free Tier meter.
- Run ./scripts/workspace.sh cost, review the fresh Terraform plan, and obtain owner approval before typing the apply phrase.
- Investigate alerts immediately. Budgets do not pause, stop, or delete resources, and billing data can be delayed.
- Recheck Billing the day after destruction because usage may appear after the resource is gone.

7. Preview and create the environment

Preview first

```
./scripts/workspace.sh plan --region us-west-2
```

Plan performs strict preflight, validates scheduling inputs when enabled, initializes the proven backend, saves workspace.tfplan, and displays the proposal. It does not create infrastructure. Read all create, update, replace, and delete lines; verify the account, Region, names, tags, instance counts, inbound CIDRs, log retention, budget recipients, and any schedule resources.

Apply only the fresh reviewed plan

```
./scripts/workspace.sh apply --region us-west-2
```

1. The helper rebuilds the plan so approval cannot accidentally apply a stale file.
2. It repeats the documented cost warning and displays the exact account/environment proposal.
3. Type I ACCEPT ESTIMATED AWS CHARGES only if the plan and estimate are approved.
4. The exact saved plan is applied. There is no general --yes switch.
5. After success, run status and open the output URL only for non-sensitive smoke testing.

```
./scripts/workspace.sh status --region us-west-2
```

CAUTION: Stop when the plan surprises you

Do not approve a plan because most lines look right. A wrong account, Region, backend, broad CIDR, unexpected replacement, unknown deletion, missing budget, or unplanned schedule is a reason to stop, correct the input, and create a new plan.

8. Optional scheduled self-destruct

STOP: Paid and time-sensitive option

Scheduling adds an AWS-hosted control plane and a dedicated U.S. toll-free two-way SMS requirement. The repository documents a rough first-month example of about US\$2.08 for one already-approved number and six single-part messages, before registration/campaign fees, retries, taxes, carrier charges, and other AWS usage. Check current pricing.

Before selecting Yes

- Deadline at least 13 hours away; duration or local YYYY-MM-DD HH:MM is converted to UTC. Ambiguous/nonexistent daylight-saving minutes are rejected.
- Operator cell in E.164 form and a monitored email address; the operator must control and consent to the notification sequence.
- A pre-existing account-owned ACTIVE U.S. TOLL_FREE AWS End User Messaging SMS number with SMS, two-way messaging, carrier-managed opt-outs, and inbound SNS in the same account/Region.
- A successful AWS no-send dry-run for the complete SMS path. The builder does not purchase or register the number.

What the operator must do after apply

1. Find the SNS subscription-confirmation email and confirm it while more than 12 hours remain.
2. Run `./scripts/workspace.sh schedule-status --region us-west-2`. PENDING means it is not armed; ACTIVE means it is.
3. Expect an ARMED notice and milestone notices at 12 hours, 2 hours, 1 hour, 15 minutes, and 5 minutes before the deadline. Scheduler/carrier/email delivery can be late.
4. To cancel, reply with the single word CANCEL to a warning SMS from the enrolled cell. Capitalization/outer spaces are ignored; extra words or punctuation are invalid.
5. Verify CANCELLED with `schedule-status`. STOP opts out of future SMS but does not cancel deletion. Email replies never cancel.

State	Meaning	Operator response
PENDING	Waiting for timely email confirmation; deletion cannot start.	Confirm while >12h remains or create a new reviewed schedule/manual destroy.
ACTIVE	Armed and enforcing deadline.	Monitor notices; reply exact CANCEL by SMS if teardown should cease.
TRIGGERING / TRIGGERED	Controller/build start is in progress; valid CANCEL may still win before execution claim.	Check status immediately; do not assume a delayed acknowledgement.
CANCELLED	Authenticated cancellation won.	Authoritative. Remove the remaining control plane later with reviewed Terraform destroy.
EXECUTING	Deletion plan has begun.	Cancellation is too late. Let dependency-ordered teardown finish; redeploy to recover.

At the deadline, CodeBuild proves the account and TRIGGERED state, downloads Terraform 1.15.8 and verifies its checksum, initializes the exact backend, creates a deletion plan, rejects any create/update action, atomically claims EXECUTING, and applies that exact plan. It removes runtime and schedule resources but keeps protected state and the first-run identity.

9. Publish and roll out the application

Build and push only after the smoke-test infrastructure exists

```
./scripts/publish_app.sh /absolute/path/to/customer-application --region us-west-2 --tag release \
-2026-07-19
```

1. The helper repeats AWS preflight and the application inspection.
2. Review the absolute source path, ECR destination, account, Region, and image tag.
3. Type BUILD AND PUSH APPLICATION IMAGE only after source review. Docker builds linux/amd64.
4. ECR returns a sha256 digest. The helper records the immutable digest in ignored terraform/application.auto.tfvars.json.
5. No instance changes yet. Run plan, inspect the launch-template/instance-refresh proposal, then apply with the cost confirmation.

```
./scripts/workspace.sh plan --region us-west-2
./scripts/workspace.sh apply --region us-west-2
./scripts/workspace.sh status --region us-west-2
```

CAUTION: One-instance refresh can be briefly unavailable

The cost-conscious default does not require two healthy instances during every rollout. Increase minimum/desired capacity to two only after approving the additional compute, storage, IPv4, and traffic cost.

Secrets and application configuration

- Never place secrets in the image, Docker build arguments, Terraform variables/state, tags, user-data, logs, or a browser-delivered front-end bundle.
- Public front-end configuration is visible to users. Private runtime configuration needs a separate Secrets Manager or Parameter Store design and a narrow instance role.
- Review ECR scan findings before promotion. This development workflow enables scanning but does not automatically block a rollout on findings.

10. Operate, observe, and update

Daily operator commands

Command	Safe meaning
<code>./scripts/workspace.sh status --region REGION</code>	Read application URL and current Auto Scaling instances.
<code>./scripts/workspace.sh logs --region REGION</code>	Follow roughly the previous hour of application logs; Control-C stops following.
<code>./scripts/workspace.sh schedule-status --region REGION</code>	Read authoritative scheduled-teardown state; does not change it.
<code>./scripts/rotate_logs.sh</code>	Apply local success/error/inventory age and count limits.
<code>./scripts/workspace.sh plan --region REGION</code>	Preview drift and proposed changes without changing AWS.

Logging and retention

- Local successful transcripts: 14 days / 20 files. Local error transcripts: 90 days / 100 files. Inventory reports: 365 days / 20 files.
- CloudWatch routine application/bootstrap logs: normally 14 days. Error and scheduled-controller/build logs: 90 days. ALB request logs: 30 days.
- Begin troubleshooting with the first ERROR in logs/errors, not the final cascade. Never paste a secret, access key, authorization header, cookie, contact value, or full sensitive request into a log or chat.

Safe update routine

1. Change one understandable concern at a time in a review branch.
2. Run local validate and application inspection.
3. Run a fresh plan and compare it with the intended change statement.
4. Obtain cost/security approval for any wider network, capacity, retention, IAM, notification, or schedule change.
5. Apply the exact plan, verify health/status/logs, and retain the error transcript if anything fails.

11. Destroy safely

Choose the correct teardown path

Path	Deletes	Keeps	Use when
Routine destroy	Terraform-managed runtime, ECR images, runtime logs, ALB logs, budgets, and schedule control plane.	Protected state bucket/history and first-run service account.	The environment is idle or will be recreated.
Scheduled destroy	Same Terraform-managed runtime and its own schedule control plane at the deadline.	Protected state and first-run service account.	A reviewed deadline should work while the workstation is off.
Full self-destruct	Runtime; optionally all backend object versions and exact bootstrap IAM shape.	Anything not explicitly proven and approved; unrelated/ambiguous assets.	The attached project is being permanently retired.

Routine destroy

```
./scripts/workspace.sh destroy --region us-west-2
```

Read the destroy plan and type DESTROY AWS WORKSPACE only when the account, environment, and resource list are correct. If Terraform partially fails, fix the first permission/dependency error and rerun the same destroy; do not delete random console resources out of order.

Full retirement always begins in review-only mode

```
./scripts/self_destruct.sh --review-only --profile cleanup-admin --region us-west-2 --project \
customer-demo --environment dev
```

- The read-only inventory is broad but cannot be universal across every service/Region. An incomplete read blocks deletion; an empty-looking list is not proof.
- Automatic deletion is narrow: Terraform state plus exact backend/user names, tags, credential shape, account, Region, and a deletion-only saved plan.
- The state bucket and service account default to KEEP. Each needs an explicit flag and additional proof. Deleting the service account requires a different cleanup identity.
- Execute is a new run with --execute and --expected-account. Its phrase is SELF DESTRUCT ACCOUNT PROJECT ENVIRONMENT. Never guess it or automate around it.
- Broad account inventory never authorizes deletion of unrelated or ambiguous assets.

STOP: Permanent means permanent

State versions, images, logs, budget definitions, and IAM credentials may be irrecoverable.

Export required records under an approved retention process and obtain independent approval between review and execute.

12. Troubleshooting and recovery

Symptom	Required response
Missing tool / old AWS CLI	Stop. Install/update from the vendor instructions in docs/WORKSTATION_SETUP.md; reopen terminal; rerun validate/preflight.
No/expired credentials	Do not request raw keys in chat. Use the approved aws login or aws sso login profile, then prove with aws sts get-caller-identity.
Wrong account or root during normal work	Stop. End root, select the saved service/approved profile, and rerun STS/preflight. --run-as-root is not a convenience bypass.
First run partly failed	Read the retained error log. Reconcile any tagged IAM user/policy/key/profile created by that run before rerunning.
IAM simulation/permission failure	Have an administrator review docs/PERMISSIONS.md, organization guardrails, quotas, and exact failed actions. Do not weaken the plan to silence the check.
Missing backend.hcl / S3 AccessDenied	Run guarded backend bootstrap if absent; otherwise confirm profile, account, Region, deterministic bucket, key, and policy.
Terraform lock	Confirm the other operator/automation has stopped. Do not remove the lock blindly.
App inspection fails	Correct Dockerfile/.dockerignore, lock files, secret exposure, listener, port, or health contract. Rerun inspection before Docker receives the directory.
Targets unhealthy	Check error logs first; confirm 0.0.0.0:port, 200-399 health path, ECR digest/amd64, then use Session Manager if logs are insufficient. Do not open SSH.
Schedule PENDING	Confirm SNS email with >12h remaining. If late, create a new reviewed schedule or manual destroy; never edit DynamoDB to force ACTIVE.
CANCEL has no acknowledgement	Read schedule-status. CANCELLED is authoritative. Otherwise check enrolled source phone, destination AWS number, exact word, and whether state already reached EXECUTING.
Partial destroy	Keep backend and identity for recovery. Fix first error, rerun dependency-ordered Terraform/manual ownership-safe process, then inventory leftovers.

13. Security, data, and team handoff

Non-negotiable handling rules

- Do not share AWS secret keys, session tokens, phone numbers, email addresses, tfstate, backend credentials, .env files, or sensitive logs in Git, tickets, chat, screenshots, or source packages.
- Treat anyone who can read Terraform state as able to read scheduled contact values. State is encrypted/versioned, but not magically secret from authorized readers.
- Use Session Manager instead of SSH. IMDSv2 and encrypted root volumes are required. Instances receive only their application runtime role, never deployment credentials.
- Keep application data outside the instance. This starter kit has no backup policy for application content because it assumes the content is disposable.
- Review public HTTP exposure and allowed CIDRs. Add TLS and an application-specific secrets/data design before the system handles sensitive information.

Handoff packet for the next operator

Record	What to hand over safely
Ownership	Account owner, cost owner, application owner, approved 12-digit account ID, Region, project, environment.
Source	Builder version/checksum, destination path, reviewed Git commit/PR, local customizations.
Application	Image digest, port, health path, stateless-data design, build/release instructions.
Access	Approved profile/role name and login method - never the credential secret itself.
State	Backend bucket/key/Region and retention owner - no copied tfstate.
Cost	Estimate date, budget recipients/confirmation status, Free Tier alert setting, approved maximum capacity.
Schedule	Enabled/disabled, UTC and local deadline, state, channel owner, cancellation instructions - redact contact values in broad handoff.
Recovery	Last successful validate/plan/apply/status, known errors, retained log path, intended destroy date.

14. Command card and acceptance checklist

First complete run

```
# In the attached infrastructure/aws-envbuilder folder
./scripts/workspace.sh validate

# One time only, under verified AWS account root
aws login --profile aws-root-bootstrap
./scripts/first_run_setup.sh --root-profile aws-root-bootstrap --region us-west-2
aws logout --profile aws-root-bootstrap

# Normal service-account operation
cp terraform/terraform.tfvars.example terraform/terraform.tfvars
./scripts/bootstrap_backend.sh --region us-west-2 --project customer-demo --environment dev
./scripts/workspace.sh preflight --region us-west-2
./scripts/inspect_app.sh /absolute/path/to/customer-application
./scripts/workspace.sh plan --region us-west-2
./scripts/workspace.sh apply --region us-west-2
./scripts/workspace.sh status --region us-west-2

# Publish and roll out the actual application
./scripts/publish_app.sh /absolute/path/to/customer-application --region us-west-2 --tag VERSION
./scripts/workspace.sh plan --region us-west-2
./scripts/workspace.sh apply --region us-west-2

# Normal end of use
./scripts/workspace.sh destroy --region us-west-2
```

Acceptance checklist

- [] Package checksum verified; extracted contents reviewed; no local/state/credential artifacts present.
- [] Application inspection passes; listener/port/health/stateless/logging requirements demonstrated.
- [] STS account/identity correct; root used only for successful first run; service profile passes strict preflight.
- [] terraform.tfvars and backend.hcl are ignored/local, correct, and owned by this environment.
- [] Cost owner approves estimate, budget recipients, Free Tier alerts, instance maximum, retention, and optional schedule fees.
- [] Plan contains only intended actions; exact apply confirmation entered by an authorized person.
- [] Status/health/logs verified; image is pinned by digest; no sensitive data crosses default HTTP.
- [] Destroy date/owner recorded; routine or scheduled/full path selected; post-destroy inventory and next-day Billing check assigned.

15. References and definitions

Repository guides

- README.md - complete command syntax and safe path.
- docs/PACKAGING.md - source archive contents, checksum, attachment, and upgrades.
- docs/APP_INTEGRATION.md - Docker/health/stateless application contract.
- docs/FIRST_RUN.md and docs/PERMISSIONS.md - AWS root handoff and audited action set.
- docs/COSTS.md - estimates, Budgets behavior, and official pricing links.
- docs/SCHEDULED_SELF_DESTRUCT.md - channel requirements, deadline, state machine, notices, and cancellation.
- docs/SELF_DESTRUCT.md - inventory, ownership proofs, review/execute split, and optional permanent scopes.
- docs/TROUBLESHOOTING.md - retained logs and recovery routes.

Official AWS references

AWS CLI sign-in: <https://docs.aws.amazon.com/cli/latest/userguide/cli-configure-sign-in.html>

IAM access-key security: https://docs.aws.amazon.com/IAM/latest/UserGuide/securing_access-keys.html

AWS Free Tier alerts: <https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/tracking-free-tier-usage.html>

AWS pricing calculator: <https://calculator.aws/>

End User Messaging pricing: <https://aws.amazon.com/end-user-messaging/pricing/>

Two-way SMS setup: <https://docs.aws.amazon.com/sms-voice/latest/userguide/two-way-sms.html>

Required opt-out keywords: <https://docs.aws.amazon.com/sms-voice/latest/userguide/keywords-required.html>

EventBridge pricing: <https://aws.amazon.com/eventbridge/pricing/>

CodeBuild pricing: <https://aws.amazon.com/codebuild/pricing/>

Definitions

Term	Meaning in this kit
AWS root	The account root identity whose STS ARN ends in :root; unrelated to sudo or OS root.
Service account	The dedicated IAM user/profile created once for requested standalone operation; organizations should prefer a short-lived deployment role.
Stateless	The application remains correct when any instance and its local disk disappear.
Plan	A saved proposal. It changes nothing until the exact reviewed plan is applied.
Routine destroy	Terraform-managed runtime removal that keeps protected backend and first-run identity.
Full self-destruct	Separate inventory/proof/approval workflow that may also remove state versions and the exact bootstrap identity.
Fail closed	When a prerequisite cannot be proved, the operation stops instead of guessing or weakening the safeguard.